POST-ADSORPTION PHAGE RESISTANCE

Does intracellular defense explain infection failure once surface compatibility and clonal lineage are held fixed?

30 scripts

38,435 strain-phage pairs

INPUT DATA

Interaction matrix
402 Escherichia strains against 96 phages
38,435 observed pairs, plaque score 0 to 4
Gaborieau et al., Nature Microbiology 2024

Host genotypes
137 DefenseFinder systems per strain
LPS core, O-antigen, capsule, 12 OMP loci
core-genome tree and distance matrix

Phage genomes and models
96 genomes, VIRIDIC tree, RBP assignments
178 anti-defense profile models
7 lysogeny profiles for lifestyle calls

Bacterial genomes
287 retrieved from NCBI one strain at a time
254 pass an independent identity check
not redistributed; digests are tracked

Public inputs, verified on every run
15 files with SHA-256 digests tracked in the repository. The data itself is re-downloaded rather than redistributed.

ANALYSIS PHASES

01 — Acquire and validate
Verify digests, assert the observed dimensions, and audit the released matrix against 318,816 replicate spot tests: the binary lysis call is reproduced 99.69 percent of the time.

02 — Separate the two gates
Define receptor compatibility from isolation hosts, then stratify rather than filter. The gate admits 65.7 percent of pairs and enriches lysis only 1.34 fold, so filtering would discard signal.

03 — Associate, then remove the lineage
Condition on phage identity and surface type, refit with errors clustered on clonal background, and partition the variance by exact Shapley value over all orderings of the blocks.

04 — Stress-test whatever survives
Within-cluster fits, permutation inside a lineage, minimum detectable effects, E-values, genus heterogeneity, matched phylogenetic nulls, and full Bayesian pooling across 79 systems.

05 — Search beyond the catalogue
Test every variable k-mer across 254 genomes, 10,351 loci, against an empirical null with a locally fitted tail, then relocate and annotate whatever it flags.

06 — Resolve the panel into phages
Score each system against each phage separately, counting distinct phage groups, against a coordinate-matched and a clonally-matched null, and report the worse of the two p-values.

OUTPUTS

Result tables
58 CSV files
every number behind every figure

Plain-text reports
23 reports
each stage explains its own result

Manuscript figures
11 figures
PNG and PDF, byte-reproducible

Ranked candidate list
11 systems nominated
ordered by independent checks passed

What the analysis concludes
Of 33 testable systems, almost every association is a property of the bacterial lineage rather than of the system. No system passes all seven independent checks, and roughly three quarters of infections still fail after accounting for surface type, defense carriage and prophage content.

Reproducible
121 outputs regenerate byte for byte

Verified
226 claims checked against the tables

Tested
estimators checked against statsmodels

Confounder-aware
surface, lineage and island adjusted

Nulls reported
negative results carry equal weight